Benedict Armstrong

ML for physics · Computational Science & Engineering, ETH Zürich

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Computational Science and Engineering student at ETH Zürich working at the intersection of machine learning and physics. My research focuses on sequence models for detecting and characterising gravitational-wave signals. I also have several years of experience shipping full-stack software as a freelancer and have taught algorithms, data structures and software engineering to hundreds of students.

EDUCATION

2026 – present	**M.Sc. Computational Science and Engineering** · ETH Zürich	Robotics track
2021 – 2025	**B.Sc. Computational Science and Engineering** · ETH Zürich	Robotics track
2019 – 2021 (switch to ETH)	**B.Sc. Computer Science** · Vienna University of Technology	Software & Information Engineering track

RESEARCH

AI for Gravitational-Wave Physics

June 2025 – Summer 2026

ML Research Intern, Computational Applied Mathematics & AI Lab (CAMAIL), ELLIS Institute Tübingen and Max Planck Institute for Intelligent Systems

In collaboration with the LIGO Lab at MIT · supervised by T. Konstantin Rusch

When two neutron stars spiral into each other, they send ripples through spacetime that LIGO measures as length changes far smaller than a proton. These signals can last minutes and are buried in non-stationary, glitchy detector noise, while astronomers need an alert within seconds to point telescopes at the merger. I work on Linear Oscillatory State-Space Models (LinOSS): sequence models that handle very long inputs efficiently and make it possible to detect binary neutron star mergers in real time.

- Designed and trained state-space models for low-latency binary neutron star detection on LIGO strain data.
- Built the training and evaluation pipeline in PyTorch (Lightning) and JAX/Equinox.
- Co-developed PhyTS, a benchmark of physics time-series tasks (NeurIPS 2026).

TALKS

Linear Oscillatory State-Space Models for Binary Neutron Star Detection — Benedict Armstrong, *AI for Gravitational Waves Workshop*, CERN, Geneva, May 2026

Direct Parameter Estimation of Binary Neutron Star Mergers with State Space Models — Kevin (Kyungseop) Yoon, *Fast Machine Learning for Science Conference 2026*, UC San Diego, September 2026

Python PyTorch PyTorch Lightning JAX Equinox

PUBLICATIONS

2026

PhyTS: A Benchmark for Scientific Time Series

Benedict Armstrong, Jeroen Audenaert, Hannah P. Binney, Alice Cheng, Maarten De Vos, Luis F. Domingues, ..., Kyungseop Yoon

Advances in Neural Information Processing Systems (NeurIPS) (NeurIPS 2026), Sydney

EXPERIENCE

2022 – 2024

Freelance Software Engineer · Armstrong Systems

Founded Armstrong Systems, software consulting for clients in Zürich. Full-stack development of various projects including a web-based data management tool, visualization dashboards and large-scale data ingestion pipelines.

- Scoped, built and maintained a web-based data management tool used day to day by the client's team.
- Built interactive visualization dashboards on top of the clients' operational data.
- Designed large-scale data ingestion pipelines, from source integration to deployment and monitoring.

Fall 2024

Coding Assistant for LehrFEM++ Finite Element Library · Seminar for Applied Mathematics, ETH Zürich
Coding assistant for the C++ finite element library LehrFEM++, used in the course *Numerical Methods for Partial Differential Equations* taught by Prof. Dr. Ralf Hiptmair.
- Extended and maintained the LehrFEM++ codebase used by students in the course's programming exercises.
- Wrote documentation and examples to make the library easier for students to pick up.

2018 – 2019

Paramedic (civil service) · Samariterbund
Worked as a paramedic in Vienna as part of my civil service.

Summers 2017–2021

Various summer jobs · Various
Various summer jobs in Vienna, including working as a postman, food delivery driver and math/CS tutor for high school students.

TEACHING

Department of Computer Science, ETH Zürich

Spring Semester 2024

Teaching Assistant · *Software Engineering Fundamentals*
- Led a weekly exercise session and gave feedback on students' Python projects.

August 2023 – February 2024

Head Teaching Assistant · *Computer Science II*
- Coordinated the team of teaching assistants and the weekly exercise classes.
- Prepared exercise material together with the lecturers.

August 2022 – February 2023

Teaching Assistant · *Computer Science II*
- Taught a weekly exercise class and reviewed student submissions.

Research Group Industrial Software (INSO), TU Wien

Spring 2021

Teaching Assistant and Full Stack Developer · *Security for Systems Engineering*
- Developed security challenges for a student Capture the Flag (CTF) competition with a small team.
- Supervised students during the competition and helped administer exams.

PROJECTS

Presio presio.ch

Web app for presenting PDF slides with a presenter controller and a mirrored audience window. PDFs stay in the browser by default; signed-in users can share a session code so others can follow live on their own devices.

AcceptOdds acceptodds.com

Short description of what AcceptOdds does, who it is for and what is technically interesting about it.

SKILLS

Languages: English, German

Technologies: Python, PyTorch, JAX, C++, C, Java, JavaScript, PHP, SQL, Angular, React, NextJS, Svelte(Kit), FastAPI, REST, HTML, CSS, Mongo, Postgres, Docker, Git, Arduino/ESP8266/32 based hardware development, Scrum